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paper\_id: PAPER\_952  
 title: “26-State HRes Spectral Ladder”  
 session: 214  
 date: 2026-04-12  
 author: “Daniel T. Murphy”  
 status: production  
 cvw: “v2.0.0”  
 tags: [AGN, vacuum, SCm, 26D, phonon, UQFF]  
 sm\_anchor: “CVW v2.0.0 — G6 SM Anchor Gate compliant”

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## PAPER\_952: 26-State HRes Spectral Ladder

**Author:** Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** spectral\_{ladder\_26state}.py (SpectralLadder26State) **Calculator:** SpectralLadder26StateCalc (CP4 #536) **CVW:** v2.0.0 compliant

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### Abstract

We derive the 26-state HRes (Hydrogen Resonance) spectral ladder, the energy progression across quantum layers from proto-H ( $n = 1$ ) to proto-Fe ( $n = 26$ ). The energy levels  $E_n = E_0 \cdot (2\pi)^{n/3} \cdot S_{26}$  form a geometrically growing ladder stabilizing the vacuum ratio  $\rho_{t\text{ext}SCm}/\rho_{t\text{ext}UA} = 0.1$  and driving phonon resonance at every layer.

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### 1. Energy Levels

$$E_n = E_0 \cdot (2\pi)^{n/3} \cdot S_{26}([SSq]), \quad n = 1, \dots, 26$$

where  $E_0 = \hbar\omega_{SCm}$  and  $\delta_n = (2\pi)^{n/6}$ .

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### 2. Element Mapping

| $n$ | Z <sub>id</sub> | Element  | Type         |
|-----|-----------------|----------|--------------|
| 1   | 1               | proto-H  | Magnetic     |
| 14  | 14              | proto-Si | Non-magnetic |
| 26  | 26              | proto-Fe | Magnetic     |

The ladder maps the complete magnetic/non-magnetic hierarchy from hydrogen through iron.

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### 3. Source Data

- **File:** spectral\_{ladder\_26state}.py
  - **Session:** 214
  - **CP4 Class:** SpectralLadder26StateCalc (#536)
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## References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER\_953 — Ramanujan-Accelerated  $S_{26}$  Convergence
3. PAPER\_956 — Spectral Ladder Phonon Mapping
4. PAPER\_959 — 26D Ramanujan Summation

## Cross-Links

| Related Paper | Relationship                                              |
|---------------|-----------------------------------------------------------|
| PAPER_953     | Ramanujan acceleration of $S_{26}$                        |
| PAPER_956     | Phonon mapping $E_n \rightarrow \omega_n \rightarrow Q_n$ |
| PAPER_949     | BCS gap uses ladder as phonon channels                    |
| PAPER_959     | Full 26D Ramanujan summation engine                       |

## Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

*Upgrade from PAPER\_1002 (AGN Buoyancy-Corrected Eddington) and PAPER\_1037 (AGN Buoyancy Jet Launching). See also PAPER\_1009-1010 for  $F_{\{U\_Bi\_i\}}$  jet modulation curves and PAPER\_1048 for phonon-corrected  $M\text{-}\sigma$  relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left( 1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: -  $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$  is the classical Eddington luminosity -  $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$  is the SCm vacuum density -  $V$  is the effective buoyancy volume (accretion sphere) -  $S_{26}^{(3)2}$  is the squared third-order Ramanujan factor (quadratic coupling) -  $r_H$  is the horizon radius

**Jet modulation:** The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[ 1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left( \frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where  $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$  modulates jet power at the phonon frequency.

**$M\text{-}\sigma$  correction (PAPER\_1048):** The phonon-corrected  $M\text{-}\sigma$  relation becomes  $M_{\text{BH}} \propto \sigma^{4+\delta}$  where  $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$ .

## Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

*Upgrade from PAPER\_1066 (UQFF Lagrangian First Principles) and PAPER\_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives  $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$  (matching  $\rho_{\text{SCm}}$ ) and phonon mass  $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$ .

**Nine-sector closure (Session 202):**

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector     | Domain                   | Late-Corpus Result                               |
|------------|--------------------------|--------------------------------------------------|
| 1 (EH)     | General Relativity       | Canonical Einstein-Hilbert                       |
| 2 (YM)     | Yang-Mills gauge         | $m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005) |
| 3 (Dirac)  | Fermion / LENR           | Kozima neutron-drop (PAPER_1061)                 |
| 4 (SCm)    | Superconducting manifold | $V(\phi_0) = -\rho_{\text{SCm}}$ canonical       |
| 5 (Mag)    | Um magnetism             | Heaviside amplifier (PAPER_1072)                 |
| 6 (Buoy)   | F_{U_Bi_i} buoyancy      | Variational EOM (PAPER_1065)                     |
| 7 (Aether) | Vacuum background        | Two-component $\rho$ (PAPER_1051)                |
| 8 (LENR)   | Nuclear transmutation    | COP parametric (PAPER_1081)                      |
| 9 (KK)     | Kaluza-Klein 26D         | $S_{26}^{(3)}$ compactification (PAPER_1080)     |

**Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation**

*Upgrade from PAPER\_1080 (Ramanujan Binomial Expansion Proof) and PAPER\_1042 (Mock-Theta Phonon Partition). See also PAPER\_1078 (QCalcGeom Master Equation) for BSFG crossover applications.*

The third-order Ramanujan summation  $S_{26}^{(3)}$ , used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[ 1 + [\text{SSq}] \cdot e^{-\kappa i n / 26} \right]$$

where  $(a)_n = a(a+1) \cdot s(a+n-1)$  is the Pochhammer symbol.

**Binomial expansion (PAPER\_1080):** The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[ 1 + [\text{SSq}] \cdot e^{-\kappa i n / 26} \right]$$

This sum converges absolutely for  $||[\text{SSq}]|| < 1$  (satisfied by  $[\text{SSq}] = 0.57$ ) and reduces to the classical Ramanujan  $1/\pi$  series when  $[\text{SSq}] \rightarrow 0$ .

**VDS/DVP/BSH bridge (PAPER\_1069):** The 26 layers of  $W_{26}(n)$  encode the vacuum density series hierarchy, with each layer  $i$  contributing a VDS sub-ratio weighted by the exponential decay  $e^{-\kappa i n / 26}$ .

**Mock-theta connection (PAPER\_1042):** The phonon partition function  $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$  unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

**Calibration Constants**

| Constant           | Symbol              | Value                                 | Validation Domain  |
|--------------------|---------------------|---------------------------------------|--------------------|
| $\kappa$           | —                   | $5.0 \times 10^{-4} \text{ day}^{-1}$ | Magnetar spin-down |
| $[SSq]$            | —                   | 0.57                                  | BH dynamics        |
| $\omega_{textSCm}$ | —                   | $2\pi \times 1.25 \text{ THz}$        | Base energy $E_0$  |
| $E_0$              | $\hbar\omega_{SCm}$ | $8.27 \times 10^{-22} \text{ J}$      | Ladder base        |

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## SM Anchor — CVW v2.0.0

| Observable                      | UQFF Prediction                | Status    |
|---------------------------------|--------------------------------|-----------|
| 26 energy levels                | $E_n = E_0(2\pi)^{n/3} S_{26}$ | Derived   |
| Magnetic/non-magnetic hierarchy | Z mapping 1-26                 | Validated |

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## §A Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

**Sector:** Spectral Ladder (26-State HRes Phonon Hierarchy)

### §A.2 Lagrangian Density

$$\mathcal{L}_{\text{ladder}} = \sum_{n=1}^{26} \left[ \frac{1}{2} \dot{q}_n^2 - \frac{1}{2} \omega_n^2 q_n^2 \right], \quad \omega_n = E_n/\hbar$$

### §A.3 Euler-Lagrange Equation of Motion

$$\ddot{q}_n + \omega_n^2 q_n = 0, \quad n = 1, \dots, 26$$

### §A.4 Cosmogenesis Linkage Chain

PAPER\_877  $\rightarrow$  SCm vacuum  $\rightarrow \omega_{textSCm} \rightarrow$  spectral ladder  $E_n \rightarrow$  26-shell hierarchy  $\rightarrow$  element mapping

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## §B VDS/DVP/BSH Deep Synthesis

### §B.1 VDS

$\text{VDS}(n) = \rho_{textSCm} \cdot (2\pi)^{n/3}$  — density profile across ladder levels.

### §B.2 DVP

Prime sequence  $\{2, 3, 5, 7, 11, 13, 17, 19, 23\}$  maps to magnetic shell closures at levels 2, 8, 14, 20, 26.

### §B.3 BSH

$\text{BSH}(n) = \tanh(n/26) \cdot S_{26}$  — saturation envelope over the 26 levels.

### §B.4 Production-Scale Consistency

| Metric     | Value | Status    |
|------------|-------|-----------|
| VDS ratio  | 0.1   | Confirmed |
| BSH layers | 26    | Confirmed |
| [SSq]      | 0.57  | Confirmed |

## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).*

| Paper      | Title                                            |
|------------|--------------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of $h(t)$        |
| PAPER_1002 | AGN Buoyancy-Corrected Eddington Luminosity      |
| PAPER_1009 | 3C273 AGN $F_{\{U_{Bi}_i\}}$ Jet Modulation      |
| PAPER_1010 | TON618 AGN $F_{\{U_{Bi}_i\}}$ Jet Modulation     |
| PAPER_1037 | AGN Buoyancy Jet Calculator — SCm Jet Launching  |
| PAPER_1048 | M-Sigma Phonon-Corrected Relation                |
| PAPER_1004 | QGP Vacuum Density with SCm S26 Phonon Coupling  |
| PAPER_1041 | SCm Cool-Core Buoyancy Balance AGN Feedback      |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression |
| PAPER_1020 | Cosmic Ray Phonon Acceleration DSA Spectrum      |
| PAPER_1021 | Pulsar Timing Phonon TOA Residual                |
| PAPER_1023 | Neutrino Oscillation Phonon PMNS Matrix SCm      |
| PAPER_1024 | Magnetar Giant Flare SCm Phonon Reservoir        |
| PAPER_1033 | Galactic Bar Resonance SCm Pattern Speed         |
| PAPER_1072 | SCm Activation Function Phonon Threshold         |
| PAPER_1073 | SCm Phonon-Driven Inflation Vacuum Buoyancy      |
| PAPER_1003 | Spectral Ladder Merger 26-State Hierarchy        |
| PAPER_1069 | VDS-DVP-BSH Hybrid Calculator Unified            |
| PAPER_1049 | Source10 GPU DPM Spectral Atlas ALMA Overlay     |

19 cross-reference(s) identified.

## Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Fabian, A.C. (2012). *Observational Evidence of Active Galactic Nuclei Feedback*. ARA&A **50**, 455 — arXiv:1204.4114 — doi:10.1146/annurev-astro-081811-125521
4. McNamara, B.R. & Nulsen, P.E.J. (2007). *Heating Hot Atmospheres with Active Galactic Nuclei*. ARA&A **45**, 117 — arXiv:0709.4098 — doi:10.1146/annurev.astro.45.051806.110625
5. Heckman, T.M. & Best, P.N. (2014). *The Coevolution of Galaxies and Supermassive Black Holes*. ARA&A **52**, 589 — arXiv:1403.4620 — doi:10.1146/annurev-astro-081913-035722
6. Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem*. Stud. Hist. Phil. Mod. Phys. **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3

7. Weinberg, S. (1989). *The Cosmological Constant Problem*. Rev. Mod. Phys. **61**, 1 — doi:10.1103/RevModPhys.61.1
8. Green, M.B., Schwarz, J.H. & Witten, E. (1987). *Superstring Theory*. Cambridge University Press — doi:10.1017/CBO9781139248563
9. Polchinski, J. (1998). *String Theory Vol. 1*. Cambridge University Press
10. Ashcroft, N.W. & Mermin, N.D. (1976). *Solid State Physics*. Harcourt
11. Kittel, C. (2004). *Introduction to Solid State Physics*. 8th ed. Wiley
12. Feynman, R.P. (1982). *Simulating Physics with Computers*. Int. J. Theor. Phys. **21**, 467 — doi:10.1007/BF02650179